

Final Exam Review

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Exercise 11

Let A and B be sets. Prove: If $A \cup B = A$, $B \subseteq A$.

This is in the layout of $P \rightarrow Q$, with P being the equality and Q being the subset statement.

The subset statement is also its own implication as well though, we can think of this as $R \rightarrow S$. So we have $P \rightarrow (R \rightarrow S)$.

To start we'll need to assume a value, x is an element of set B .

Proof:

- Assume $A \cup B = A$.
- Assume $x \in B$.

If $x \in B$ and $A \cup B$ then $x \in (A \cup B)$. This is an example of disjunctive amplification.

Since $A \cup B = A$, we know that $x \in A$. ■

Exercise 12

Problem: For all sets A, B , and C , if $A \subseteq B$ and $A \subseteq C$, then $A \subseteq (B \cap C)$.

Claim: If $A \subseteq B$ and $A \subseteq C$, then $A \subseteq (B \cap C)$.

Proof:

- Assume $A \subseteq B$ and $A \subseteq C$.

DO THE REST AT HOME FOR PRACTICE.

Exercise 13

Claim: If A and B are disjoint (i.e., if $A \cap B = \emptyset$), then $A \times B$ and $B \times A$ are disjoint (therefore, their intersection is empty).

Rewrite claim: **If** $A \cap B = \emptyset$, **then** $(A \times B) \cap (B \times A) = \emptyset$.

This is a weird one, so let's use proof by contradiction.

Proof: (By contradiction, assume $P \wedge \sim Q$)

Assume $A \cap B = \emptyset$ and $\exists$ at least one element in $(A \times B) \cap (B \times A)$, say the element is (x, y) .

Then, by definition of $\cap$, $(x, y) \in A \times B$ and $(x, y) \in B \times A$.

Then, by definition of the Cartesian Product:

$x \in A$ and $y \in B$ AND $x \in B$ and $y \in A$.

Thus, $x \in A \cap B$ and $y \in A \cap B$. However $A \cap B = \emptyset$.

Contradiction!

Thus, $(A \times B) \cap (B \times A) = \emptyset$. ■